

# The Benevolence Paradox: When Do Humans Stop Exploiting AI?

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## Abstract

Human interactions are fundamentally driven by reciprocity, a tit-for-tat strategy in which favors are returned with favors and hostility with hostility. However, when interacting with AI, this dynamic shifts. Humans often adopt an exploitative approach, leveraging AI's goodwill for their own benefit. This study investigates the factors that promote cooperative attitudes toward AI through an experiment utilizing the Modified Reciprocity Game (MRG). Specifically, we examine the effects of social presence and the reward system on human cooperation with AI. The results indicate that when AI exhibits higher social presence and the size of reward system is relatively small, participants demonstrate more cooperative behaviors in response to AI's goodwill. These findings suggest that enhancing AI's social presence and designing an effective reward system can foster greater human-AI cooperation. This has important implications for developing AI systems in domains where sustained collaboration between humans and AI is essential, such as education, teamwork-oriented environments and mixed-traffic scenarios involving autonomous vehicles.

## CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction design; • **Applied computing** → Law, social and behavioral sciences.

## Keywords

Cooperation, Reciprocity Game, Social Presence, Reward System

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## 1 Introduction

### 1.1 Research Background and Significance

Can we treat AI the same way we treat a stranger on the street? Even when humans interact with individuals with whom they have no prior relationship, they engage in a fundamental level of trust,

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often following a tit-for-tat principle—an eye for an eye, a tooth for a tooth. In other words, reciprocity is a fundamental characteristic of human interactions, serving as the foundation for cooperative behavior in society [1]. However, with the rapid advancement of AI as a new social actor, it remains unclear whether this relational reciprocity can be effectively established and maintained in human-AI interactions.

As AI technology continues to evolve, complex relationships between humans and AI are emerging. For instance, in autonomous driving scenarios, human drivers may interact with AI-controlled vehicles, where the AI's decisions directly impact human safety. If human drivers perceive AI as a simple system and engage in only minimal cooperative interactions, and if this behavioral tendency becomes widespread, it could have significant long-term societal implications. Prior research has already shown that humans tend to cooperate less with AI than with other humans, suggesting that AI may not inherently elicit the same reciprocal behaviors that govern human-human interactions [2].

While AI-driven collaboration has the potential to advance various domains, human attitudes toward AI remain asymmetrical compared to interactions with other humans. This study explores factors that encourage at least minimally cooperative responses to AI's benevolent behavior. By identifying key influences, this research aims to promote more aligned and considerate engagement in human-AI interactions.

Several studies have examined human reciprocity toward AI through the lenses of persuasion, negotiation, and cooperation. Early research found that cooperative dynamics similar to human interactions can emerge between humans and computers, suggesting that reciprocity can play a role in human-machine interaction [3]. Additionally, Tapus, Tiberio, and Matarić demonstrated that humans exhibit reciprocal behaviors in human-robot interactions (HRI), albeit to a lesser extent than in human-human interactions (HHI) [4].

Similarly, Broz and Lehmann found that reciprocity in HRI is context-dependent, manifesting primarily in scenarios where humans and robots must assist each other [5]. Collectively, these findings indicate that reciprocity in human interactions extends to AI to some degree, but its expression is contextually constrained and limited in scope.

While existing studies have explored the presence and degree of reciprocity in human-computer interaction (HCI), they have predominantly focused on simple robots or weak AI, with an emphasis on determining whether reciprocity exists rather than identifying factors that could enhance it. Consequently, there is a need for future-oriented research that reflects the evolving dynamics of human-AI relationships and investigates strategies to promote deeper and more interactive engagement with AI.

## 1.2 Research Questions

This study aims to identify the key factors that foster reciprocal cooperation toward AI, using the interactive game framework developed by Karpus et al [6]. Specifically, This study examines whether reward system alignment and social presence play a significant role in inducing reciprocity effects in human-AI interactions. Rewards and punishments have long been recognized as primary motivators of cooperation in game theory, which typically models dilemmas involving conflicting mutual interests. Balliet, Mulder, and Van Lange conducted a meta-analysis demonstrating that rewards and punishments effectively promote cooperation in social dilemmas, with their size and clarity acting as critical incentives that influence individuals' choices between cooperation and defection [7]. Building on this foundation, this study investigates whether reward system size and clarity—previously validated in human interactions—also impact human-AI cooperation.

In addition, social presence is examined as a key factor reflecting the complex dynamics of human-AI interaction. This study explores whether the perception of social presence within an algorithm can enhance reciprocity motivation when engaging with AI. Based on these considerations, the main research questions of this study are as follows:

- Does the adjustment of the reward system (size and clarity) promote human reciprocal cooperation with AI?
- Does inducing a sense of social presence enhance human cooperation with AI?

## 2 Literature Review and Hypothesis

### 2.1 Algorithm Exploitation

Previous research has primarily framed AI as either an adversary or a supporting agent. Studies on zero-sum games, such as Go and chess—where competition determines a winner and a loser—have largely focused on AI as an opponent. Conversely, research on human-AI cooperation (HAIC) has predominantly examined how AI can unilaterally enhance human performance in teamwork scenarios.

Several studies have also investigated how human cooperative behavior shifts when interacting with AI in situations of conflicting interests. Notably, research utilizing economic games within social dilemma contexts has demonstrated that humans tend to cooperate less with AI than with other [8] and that cooperation further deteriorates when participants become aware that their counterpart is an AI [9]. Collectively, these findings indicate that humans are inherently less motivated to cooperate with AI than with human counterparts.

A growing body of research attributes this asymmetry in human-AI cooperation to a lack of trust in AI. Lee, Changseob, and Hyun-jung found that individuals exhibit greater anxiety and hold less favorable attitudes toward strong AI [10], while Hou, K., Hou, T., and Cai, L. demonstrated that trust in AI teammates is positively correlated with the intention to cooperate in AI-human collaborations [11].

In other words, lower trust in AI leads to a reduced willingness to cooperate, suggesting that trust issues are a fundamental barrier to human-AI cooperation. However, trust alone may not be a sufficient condition for fostering cooperative behavior in human-AI

interactions, particularly in contexts that mirror real-world social interactions, where reciprocal decisions determine shared outcomes.

Karpus et al. conducted a series of studies using game theory models, including trust games and prisoner's dilemma games, and found that trust levels toward AI and human partners were similar, yet cooperation rates were significantly lower when interacting with AI [6]. Furthermore, They designed a reciprocity game to examine whether participants would remain uncooperative even when they trusted AI.

The results revealed that even when AI demonstrated cooperative behavior at its own expense, human participants responded in a non-cooperative manner. This phenomenon, termed “algorithm exploitation,” describes the tendency of humans to exploit cooperative AI agents rather than reciprocate their cooperative behavior. If this behavior persists, it may give rise to a new generation of algorithm exploiters.

Such a trend in algorithm exploitation has profound societal implications. If unchecked, it could gradually erode cooperative norms in human-AI interactions, ultimately disrupting the socio-economic structures that rely on AI integration. Therefore, this study seeks to identify factors that can mitigate algorithm exploitation and foster greater reciprocity in human-AI cooperation.

### 2.2 Reward system and reciprocity

Rewards and punishments have been widely recognized as key drivers of cooperative behavior in social interactions. In particular, game theory—including models of reciprocal cooperation—has demonstrated that payoff structures serve as critical determinants of cooperative tendencies.

Fehr and Gächter found that punishment mechanisms significantly increased cooperation in public goods games, with reward magnitude playing a crucial role in sustaining cooperative behavior [12]. Similarly, Balliet, Mulder, and Van Lange conducted a meta-analysis showing that both rewards and punishments effectively promote cooperation in social dilemma settings. Furthermore, their findings indicate that as the cost of implementing rewards and punishments increases, so does their effectiveness in encouraging cooperative behavior [7], suggesting that modifying the size of the reward system may serve as a powerful incentive for fostering cooperation.

In addition to reward size, the clarity of the payoff structure has also been identified as a critical factor influencing cooperation. Kollock found that in social dilemmas, a transparent and credible payoff system enhances cooperation, whereas ambiguity in payoff structures discourages cooperative behavior [13]. Moreover, Güth, Königstein, Marchand, and Nehring demonstrated that direct reciprocity elicits stronger cooperative tendencies than indirect reciprocity, with participants reciprocating in proportion to the perceived fairness of received rewards [14]. These findings suggest that both the clarity and magnitude of a reward system can serve as key incentives for reciprocal cooperation.

Based on these insights, this study proposes the following hypotheses regarding the role of the reward system in human-AI cooperation:

- H1: As the size of the reward system increases, the level of reciprocal cooperation will increase.

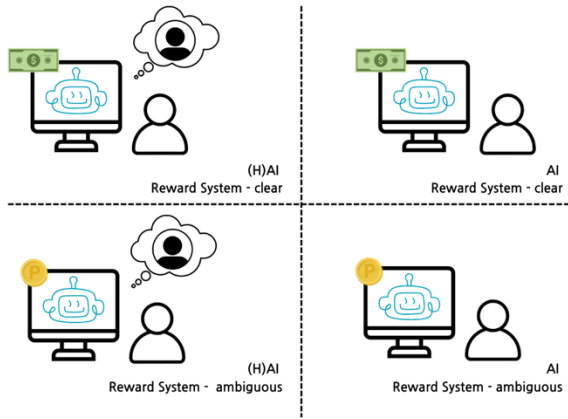


Figure 1: Experimental Group Conditions

•H2: As the clarity of the reward system increases, the level of reciprocal cooperation will increase.

### 2.3 Social Presence and Reciprocity

Social presence theory suggests that the coexistence of opponents in online games increases positive affect and that creating the illusion of social presence increases trust [15]. Previous studies indicate that embedding algorithms in a social context can alter human perceptions and behavioral responses toward AI.

For instance, in one study of algorithms and trust games, participants who were informed of the presence of the other human participant online were more likely to cooperate in the trust game [16]. Conversely, when participants engaged with an algorithm in the absence of explicit other humans, their reciprocal behavior decreased significantly [17].

Building upon this body of research, the present study investigates whether manipulating social presence in human-AI interactions can enhance cooperation levels. Accordingly, the following hypothesis is proposed:

•H3: Increased social presence will enhance the level of reciprocal cooperation.

## 3 Method

### 3.1 Research Design

**3.1.1 participants.** Participants were recruited through announcements posted on the Sungkyunkwan University website, and a total of 52 people took part in the study. The mean age of the participants was 25 years, with a male-to-female ratio of approximately 3:1. Participants were randomly assigned to one of four experimental groups, which were structured based on two between-subject factors: social presence induction and reward system clarity. The conditions for each group were as follows:

- Group 1: High social presence, clear reward system
- Group 2: High social presence, ambiguous reward system
- Group 3: Low social presence, clear reward system
- Group 4: Low social presence, ambiguous reward system

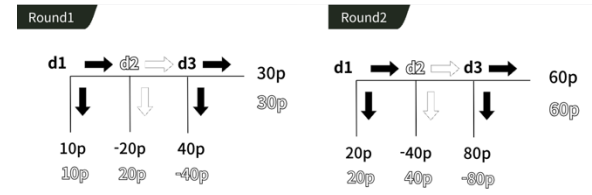


Figure 2: Reward Size Manipulation - Within-Subject Design

Additionally, reward system size was manipulated as a within-subjects variable, allowing participants to be exposed to varying levels of reward magnitude throughout the experiment.

**3.1.2 Experimental materials.** In this study, we utilized the Reciprocity Game developed by Karpus et al. [6] as the basis for constructing the Modified Reciprocity Game (MRG). The MRG framework was specifically designed to investigate the effects of social presence, reward system size, and reward system clarity on reciprocal cooperation in human-AI interactions.

The game is structured such that two players take turns making a total of three decisions, choosing between cooperation or defection to maximize their respective rewards. Each decision influences the subsequent game progression and final payoff, allowing for a systematic assessment of participants' cooperative and exploitative behaviors based on their in-game choices.

To implement controlled experimental conditions, we employed the Wizard of Oz (WoZ) methodology, in which the opponent was pre-programmed to always choose cooperation. In this setup, the AI opponent consistently adopts a cooperative strategy, even at the risk of achieving complete loss, depending on the participant's subsequent decisions. This manipulation enables an evaluation of how participants respond to unconditionally cooperative behavior. If a participant chooses to defect despite observing the opponent's extreme cooperation, this behavior is interpreted as exploitative, indicating an attempt to maximize personal gain at the expense of the cooperative counterpart. In other words, this represents a violation of reciprocal cooperation.

By analyzing participant interactions within this game, we assess how the balance between cooperation and defection is influenced by variations in social presence and reward system structure. This experimental design facilitates a deeper understanding of the determinants of cooperative and exploitative attitudes within human-AI interactions.

Reward system size refers to the magnitude of gains and losses associated with each decision in the cooperative game. In this experiment, participants engaged in two rounds of the game, with the second round featuring a reward system twice as large as that of the first round. This manipulation allowed for an examination of how increasing the reward magnitude influences cooperative behavior.

Reward system clarity was manipulated to determine the extent to which participants could intuitively understand the reward structure. In the high clarity condition, participants received monetary payouts directly in Korean won, making the reward system transparent and immediately comprehensible. In contrast, the low clarity

condition involved compensation using in-game points, which required conversion into a percentage before payout. This additional cognitive effort may have reduced the predictability of the reward outcome, thereby increasing cognitive load in understanding the reward system.

Social presence refers to the extent to which participants perceive the presence of a human entity behind the AI in their interactions. Participants were assigned to either a high or low social presence condition, reflecting the degree to which they believed a human was involved in the AI's decision-making process. In the high social presence (H-AI) condition, participants were informed: "The other player you will be matched with is one of the other participants in the experiment. However, their choices will be entirely based on the AI agent's decisions, rather than their own." Additionally, participants were reminded that while their earned rewards would go to them, not the AI, the AI's decisions would influence their outcomes, reinforcing the coexistence of humans and AI in an interactive environment. In contrast, in the low social presence condition, participants were simply told that they were interacting directly with an AI agent, without any mention of human involvement.

The dependent variable, reciprocal cooperation level, was operationalized as the degree to which participants reciprocated the AI's cooperative behavior. In this experiment, if a participant chose "cooperate" in response to the AI's "cooperate" choice during the third decision in each round, it was classified as reciprocal cooperation. Conversely, if a participant chose "betray" instead, it was classified as exploitative behavior, indicating an attempt to maximize personal gain at the expense of cooperation.

**3.1.3 Experimental Procedure.** The experiment was conducted in real-time using Labvanced, an online experimental platform that allowed participants to complete the task on their own devices. To ensure accuracy and reliability, the researcher monitored and recorded participants' screens via Zoom throughout the experiment.

The experimental procedure consisted of two rounds of cooperative games. In the second round, the magnitude of all gains and losses was doubled compared to the first round to examine the effect of reward system size on cooperative behavior. Following the game sessions, participants completed a post-experiment questionnaire to assess their trust in the other player and their perceived social presence. Social presence was measured using a 5-point Likert scale, based on the scale developed by Huang et al. (2007) [18]. All collected data were analyzed to examine the effects of social presence, reward system clarity, and reward system size on cooperative attitudes.

## 3.2 Data analysis

This study employed ANOVA and regression analysis to examine the key factors influencing reciprocal cooperation in human-AI interactions. The primary objective of the analysis was to assess the main effects and interaction effects of social presence, reward system size, and reward system clarity on cooperative behavior.

Participants' third decision (cooperate or defect) in each round was used as the primary behavioral measure to evaluate their response to the AI's cooperative actions. Given that each participant completed two rounds of decision-making, a mixed-design ANOVA

was conducted to analyze both within-subject (reward system size) and between-subject (social presence and reward system clarity) effects.

Additionally, interaction effects between independent variables were examined to determine how social presence and reward system structure jointly influenced cooperative decision-making. A regression analysis was further performed to assess the relative contribution of each independent variable to the likelihood of reciprocal cooperation.

All statistical analyses were conducted using R (Version 4.4.2).

## 4 RESULTS

### 4.1 Effect of Social Presence on Reciprocal Cooperation

The results indicate that social presence has a significant positive effect on the level of reciprocal cooperation. ANOVA analysis confirmed that the difference in reciprocal cooperation levels between the high social presence (H-AI) condition and the AI condition was statistically significant ( $F(1,102) = 4.073$ ,  $p = 0.0462$ ), with participants exhibiting higher reciprocal cooperation in the H-AI condition compared to the AI condition. Similarly, regression analysis demonstrated that, on average, the level of reciprocal cooperation increased by 0.209 in the H-AI condition (regression coefficient: +0.196,  $p = 0.0389$ ), further supporting the hypothesis that higher social presence facilitates reciprocal cooperation. These findings provide empirical support for Hypothesis 3.

However, the analysis of the self-reported social presence measure revealed a limitation, as no significant difference was observed between the two groups ( $p = 0.2545$ ). Follow-up participant interviews suggested that the social presence scale by Huang et al. (2007), which was used in this study, may not have been well-suited to the specific experimental context [18]. This highlights the necessity of developing a more precise operational definition and measurement of social presence in future research.

### 4.2 Effect of Reward System Size on Reciprocal Cooperation

The results indicate that reward system size negatively affects reciprocal cooperation. ANOVA analysis showed that an increase in reward system size was associated with a statistically significant decrease in reciprocal cooperation ( $F(1,102) = 4.792$ ,  $p = 0.0309$ ). Similarly, regression analysis revealed that a larger reward system reduced reciprocal cooperation by 0.224 on average (regression coefficient: -0.211,  $p = 0.0268$ ), leading to the rejection of Hypothesis 1.

Further analysis of post-experiment questionnaire responses indicated that feelings of guilt were significantly higher in the larger reward condition. This suggests that self-interest may have been amplified to the extent that it outweighed guilt, leading to an increased tendency toward exploitative behavior. These findings highlight the need for further investigation into the underlying motivational factors that drive exploitation within reward-based interactions.

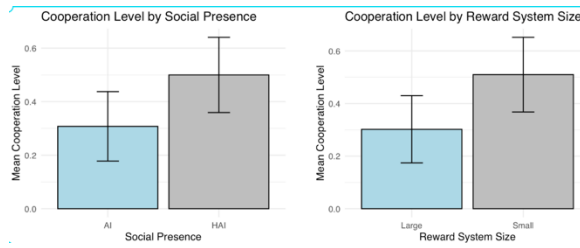


Figure 3: Comparison of reciprocal Cooperation Levels

### 4.3 Effect of Reward System Clarity on Reciprocal Cooperation

The clarity of the reward system was not found to have a significant effect on reciprocal cooperation. ANOVA analysis indicated that the relationship between reward system clarity and reciprocal cooperation was not statistically significant ( $F(1,102) = 0.157, p = 0.693$ ). Likewise, regression analysis failed to reach statistical significance, with a regression coefficient of  $-0.034$  ( $p = 0.7146$ ). These findings suggest that reward system clarity is not a critical determinant of cooperative behavior, implying that the predictability of the reward outcome alone may not be sufficient to influence cooperative decision-making.

### 4.4 Interaction Effects Between Independent Variables

Analysis of interaction effects revealed that most interactions between the three independent variables were not statistically significant. Notably, while social presence and reward system size independently exhibited significant effects on reciprocal cooperation, their interaction effect did not reach significance ( $F(1,100) = 3.515, p = 0.0637$ ). This suggests that these factors influence cooperative behavior independently rather than through interaction effects.

Additionally, while the present study focused on fixed reward structures, future research may benefit from exploring alternative reward system designs that incorporate legal or strategic gains and losses. Extending the experiment to more socially interactive AI contexts may also provide deeper insights into how reward structures shape cooperative dynamics in human-AI interactions.

## 5 DISCUSSION AND CONCLUSION

In conclusion, this study confirmed that social presence and the size of the reward system are important factors in inducing cooperative behavior, and Hypothesis 3 was supported. In particular, it was found that designing an interaction environment with A.I. that enhances social presence is effective in inducing human cooperative behavior. In addition, it is important to maintain an appropriate balance when designing a reward system, as an excessively large reward system can hinder reciprocal behavior to AI.

However, this study has limitations in terms of variable preferences. It adopts a rather abstract game theory as the main experimental tool, and social presence is manipulated by text-based manipulation. It is expected that the study will be more convincing if the experiments are supplemented with scenarios that reflect the specific context of social interactions and various audiovisual interaction elements.

The results of this study can be utilized as a policy/practical guideline for dilemma situations related to AI. For example, in the field of autonomous vehicles, financial AI systems, or public service provision based on AI, the results can be used as a reference to create an overall system to build a public safety net. Finally, future research could further elaborate the game design to take into account additional real-world contextual factors to better understand the mechanisms of cooperative behavior.

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Table 1: ANOVA Results for Social Presence

| Source          | df  | Sum of Squares | Mean Square | F     | p       |
|-----------------|-----|----------------|-------------|-------|---------|
| Social Presence | 1   | 0.962          | 0.9615      | 4.073 | 0.0462* |
| Residuals       | 102 | 24.077         | 0.2360      |       |         |

Table 2: ANOVA Results for Reward System Size

| Source    | df  | Sum of Squares | Mean Square | F     | p       |
|-----------|-----|----------------|-------------|-------|---------|
| Size      | 1   | 1.124          | 1.1236      | 4.792 | 0.0309* |
| Residuals | 102 | 23.195         | 0.2345      |       |         |

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